AI-based Collision Avoidance Lane Change System

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ABSTRACT

This study proposes a method for implementing autonomous driving functions by developing a lane change algorithm. It utilizes the YOLOv8 deep learning model and a Camera sensor to detect forward lanes and surrounding vehicles, combining LiDAR data to plan driving paths and control the vehicle. Through this, it implements autonomous driving technology that is stable and enhances the efficiency of the traffic system.

PURPOSE

The purpose of this study is to develop and validate an enhanced lane change framework that integrates deep learning–based perception with advanced Finite State Machine (FSM) control logic. The proposed framework aims to improve decision-making reliability and ensure safe trajectory generation in complex and dynamic traffic environments. By combining object detection using models such as YOLOv8 with robust state transition mechanisms, the proposed system seeks to address the limitations of conventional FSM-based approaches and enhances adaptability to real-world driving conditions. Furthermore, this research contributes to advancing the practical deployment of autonomous lane change functions in future intelligent vehicle systems.

DRIVING IMPLEMENTATION

A. Camera Sensor and LiDAR Sensor

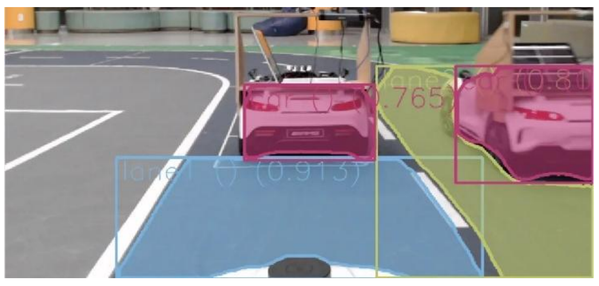

Fig 1. YOLO-Based Lane and Vehicle Detection

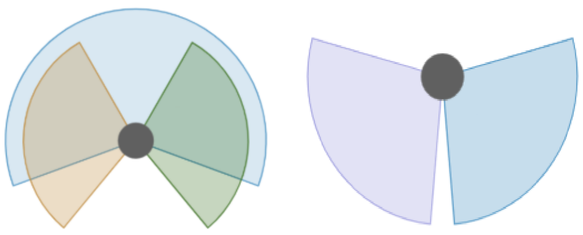

Fig 2. Detection of the LiDAR sensors

B. Pure Pursuit Path Tracking Algorithm

The Pure Pursuit path tracking algorithm sets the center of the driving lane as the target point and calculates the steering angle by determining the slope between the vehicle center and the target point. The formula for calculating is as follows:

$$\delta = \tan^{-1}\frac{2L\sin\alpha}{L_d}$$

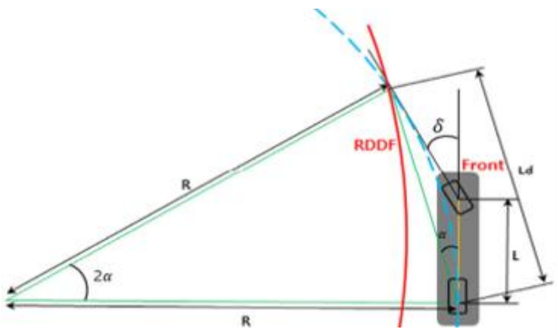

Fig 3. Bicycle Model of Pure Pursuit

C. Motor and Steering Control

a) Speed Control: The vehicle speed is primarily controlled by regulating the speeds of the left and right motors. Deceleration is determined by the distance to the vehicle ahead, and when the vehicle ahead is no longer detected, acceleration returns to the basic driving speed.

b) Steering Control: The steering angle is calculated using the Pure Pursuit algorithm, with the lane center as the reference point. To prevent abrupt steering behavior, the steering angle range is limited. When the curvature is high, a differential speed is applied between the left and right motors.

D. Lane Change Algorithm

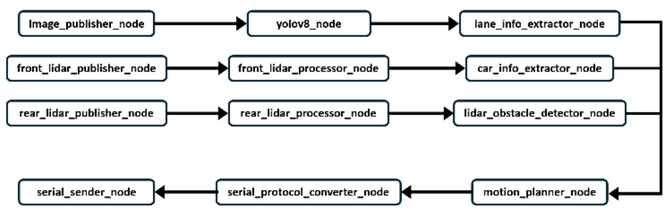

Fig 4. Overall diagram of system nodes

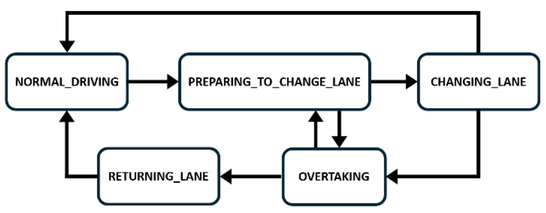

Fig 5. State diagram

• NORMAL DRIVING: This is the vehicle's basic driving state. If an vehicle in the current lane is detected via camera and LiDAR information, it transitions to the PREPARING TO CHANGE LANE state.

• PREPARING TO CHANGE LANE: The system transitions to the CHANGING LANE state once the absence of vehicles in the adjacent lane has been confirmed.

• CHANGING LANE: In this state, the vehicle actually performs the lane change. The completion of the lane change is determined by comprehensively considering the count condition and the slope of the lane.

• OVERTAKING: This state performs an overtaking maneuver. In this state, a time condition is applied to ensure the time required for overtaking. If a vehicle ahead is detected again during overtaking, it transitions back to PREPARING TO CHANGE LANE. If no vehicles are detected in the front and the lane to return to, it transitions to the RETURNING LANE state.

• RETURNING LANE: This state returns to the original driving lane. If the count condition, calculated based on vehicle speed, is met, it transitions to the NORMAL DRIVING state.

RESULTS

A. Scenario

a) Scenario 1: The vehicle ignores the vehicle ahead in the adjacent lane and continues driving. When a vehicle is detected ahead in the same lane, it performs a lane change to overtake and then returns to the original lane.

b) Scenario 2: The vehicle detects a vehicle ahead in the same lane and changes lanes to overtake. However, if another vehicle is detected ahead in the target lane, it continues overtaking that vehicle as well before returning.

c) Scenario 3: The vehicle detects a vehicle ahead in the same lane but also detects another vehicle behind in the target lane. So, it decelerates and waits. Once the rear vehicle passes, it changes lanes and then returns.

d) Scenario S: The vehicle detects a vehicle ahead in the same lane and performs a lane changing. However, during overtaking, another vehicle is detected ahead in the current lane, so it changes lanes again and treats this as a lane return, not performing another lane re-change.

B. Test Results

Each scenario was tested at least 5 times under various driving conditions, and the system's performance was evaluated using metrics such as the number of collisions and lane departures.

Overall, the system successfully performed all scenarios, even under various conditions involving changes in sunlight, lane curvature, and obstacle vehicles. The system accurately perceived lane and vehicle coordinates using the YOLOv8 model and detected obstacles through LiDAR sensors, making appropriate decisions based on FSM-based control logic.

No collisions occurred in any test, and there were two instances of minor lane departures on highly curved roads due to vehicle body limitations. These results confirm the system's collision avoidance capability.

CONCLUSIONS

This paper proposed an AI-based collision avoidance lane change system for autonomous driving. The proposed algorithm extends existing research by designing a more detailed state transition process for lane changes based on Finite State Machine (FSM). Furthermore, by combining camera and LiDAR sensors with the YOLOv8 model, it was confirmed that the system accurately recognizes road conditions and executes autonomous driving stably through subsequent decision-making. These results are expected to be effective in various road environments requiring overtaking, such as accident sites, parked vehicles, and slow-moving vehicles.

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